The Economic Impact and Market Forces of Shellfish Harvesting

Maine Fishermen's Forum
March 5th, 2026

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Gulf of Maine Research Institute

Project Goals

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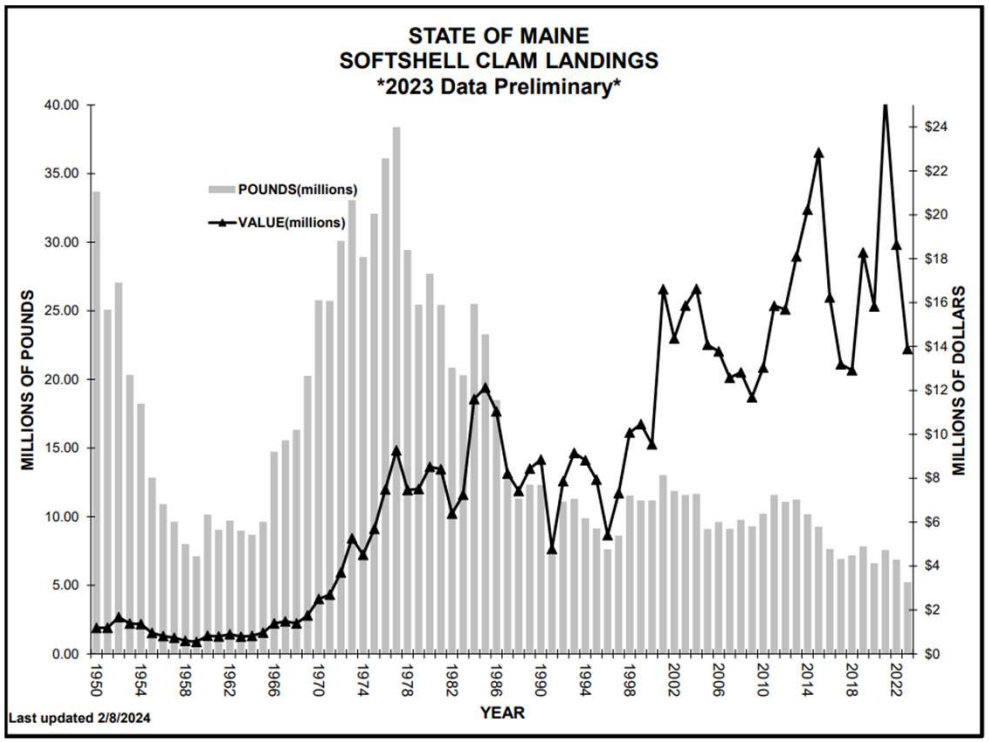
Increase understanding of landings, ex-vessel prices, and factors impacting Maine soft-shell clam fisheries and markets

- ✓ Recent trends in landed values
- ✓ Changes in market
- ✓ Factors affecting soft-shell clam landings and landed values



We'll use Fish(er) Price Index (FPI) as an approach to adjust for product-specific market inflation

Final Report

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01 – Fisheries Overview

- Soft-Shell Clam, Quahogs (Hard-shell Clam), Oysters

02 – Environmental & Socio-economic Drivers

- Environmental Factors - Tidal Heights/Sea Level, Storms, Bacterial Pollutions and Biotoxin
- Socio-economic factors - Coastal Access
- Knowledge Gaps

03 – Landings & Market Trends *(Soft-Shell Clam, Quahog)*

- Fish Price Index
- Time-Series Trends & Structural Shifts
- Price Volatility

04 – Sea Level, Storms, Closures *(Soft-Shell Clam)*

- Sea Level Impacts, Storm Impacts, Closure Impacts

05 – Synthesis Discussion

- Significant Findings
- Future Research Needs

Scan to view the final report

OR

Search "GMRI Economics Lab"

Additional Resources on GitHub

https://github.com/gulfofmaine/me-shellfish-econ-2025

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Figure jpeg files

R code

me-shellfish-econ-2025 Public

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README

me-shellfish-econ-2025

Codes for Maine DMR funded project "The Economic Impact and Market Forces of Shellfish Harvesting Understanding Landings, Ex-vessel Prices, and Factors Impacting the Soft-shell Clam Fishery and Market"

The final report is available via this link

All the figures included in the final report are available from the folder titled "DMR_TechReport_Figures".

If you have any questions, please contact Coastal and Marine Economics Lab at the Gulf of Maine Research Institute.

Scan to access GitHub repository

Physical copies at GMRI table and more at the poster session

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Notable Findings

- Landings
- Market trends and shifts
- Seasonality

2023 Landings

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- The Downeast accounted for ~50% of all landed lbs. in 2023
- Washington county reported the lowest average prices across the state ($2.27/lb)
- The Southern region reported higher average prices, likely due to their location to larger markets and demand for product
- Cumberland and York counties reported ~14% higher price per pound than Washington county

County	Number of Towns with Landings	Annual Landings (1000 lbs.)	Annual Mean Price (S.D.)
Washington	22	1,975	$2.27 (0.68)
Hancock	22	720	$2.43 (0.83)
Waldo	2	15	$2.66 (0.31)
Knox	9	347	$2.59 (0.75)
Lincoln	17	706	$2.38 (0.71)
Sagadahoc	7	259	$2.70 (0.73)
Cumberland	11	1,306	$2.60 (0.72)
York	3	35	$2.60 (0.70)

Regional Differences?

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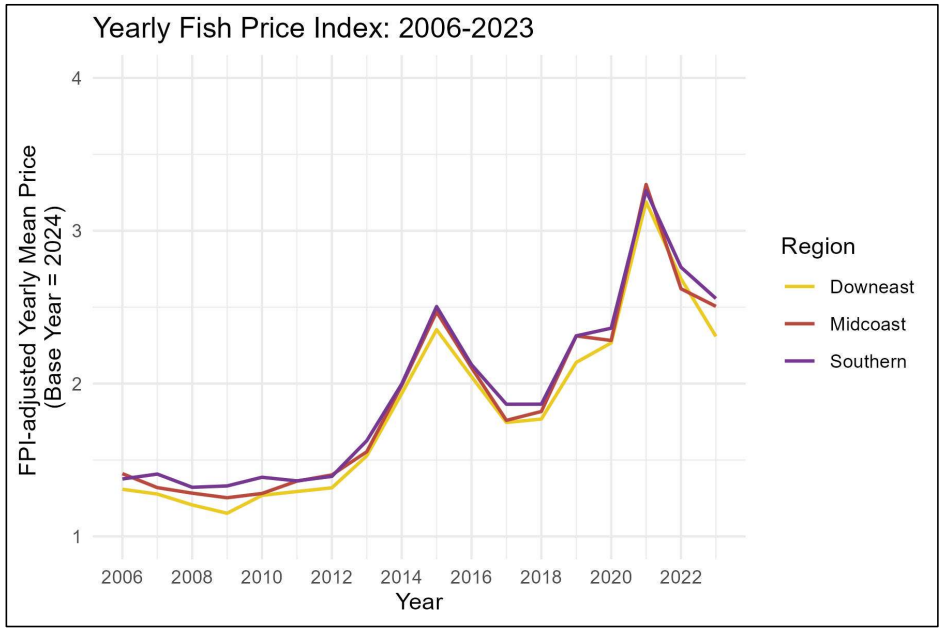
Price Movements

- Prices are higher in the Southern Maine, but prices across all regions move in the same direction

Price Jumps:

- Price spikes in 2015 and 2021
- Shifted to higher price levels since then
- Monthly average price has increased 1.8 times since 2006 ($1.43 to $2.57, FPI-adjusted)



Are Prices Jumpy?

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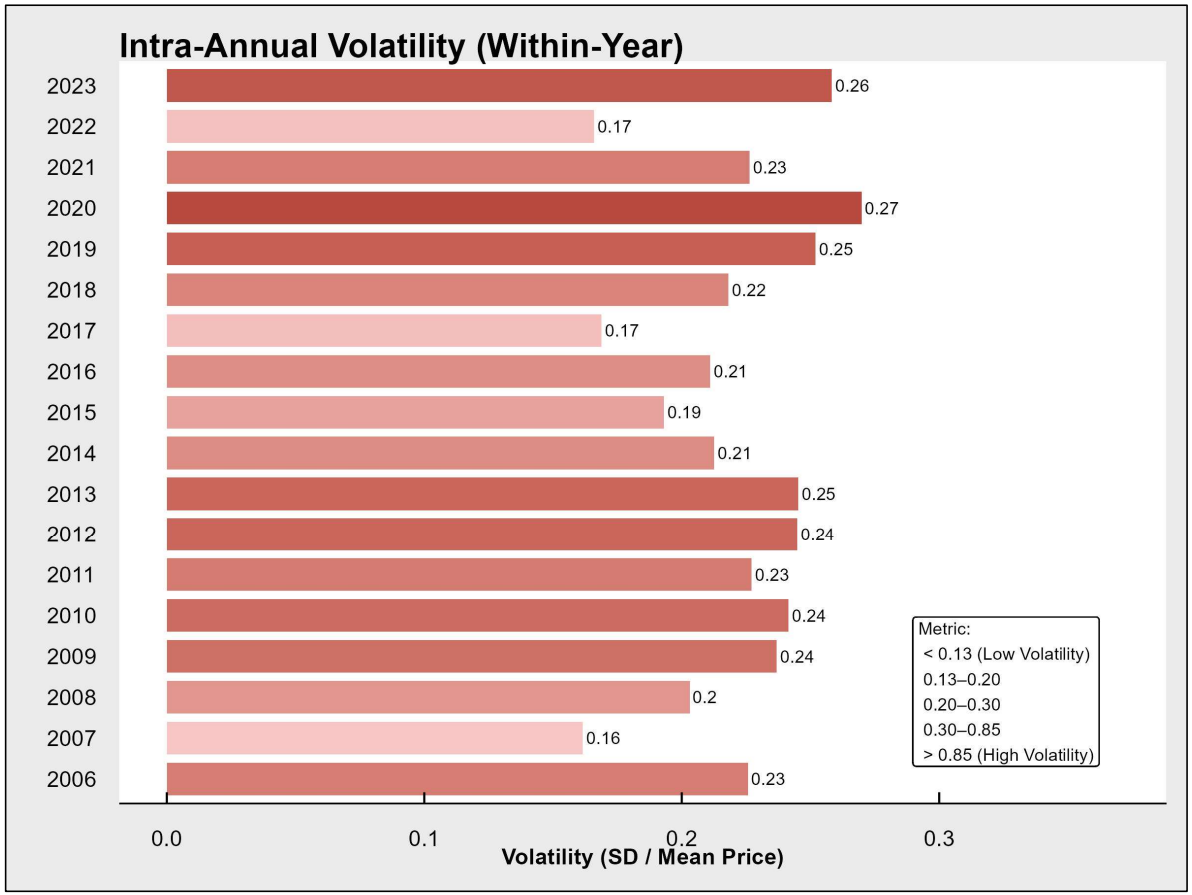
- Prices have stayed relatively stable (about the same as the groundfish prices)
- Some years are more jumpy than other years
  - 2020 ➔ COVID-19 Pandemic
  - 2023 ➔ Lowest landings recorded since 1950



Market Shifts

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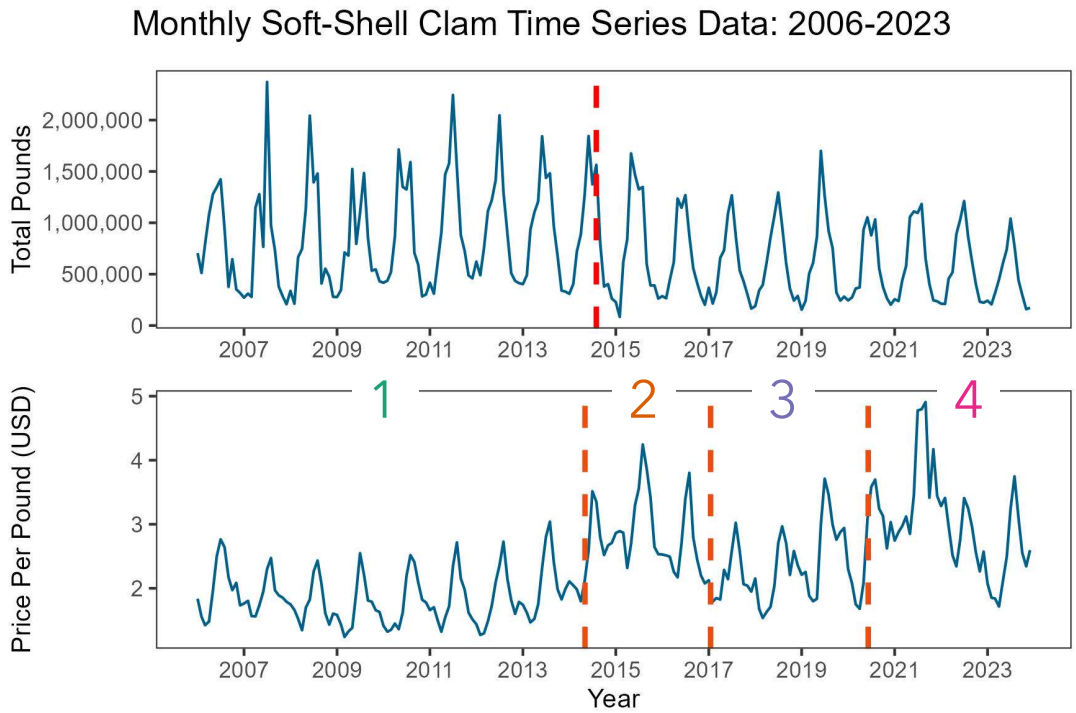
- Structural breaks in landings (volume and price)
- Possible factors
  - 2015 (first red line): Environmental effects shift landings lower and prices higher
  - 2020 (third red line): COVID-19 pandemic shifts prices higher



Something has caused fundamental shifts in landings market

Market regimes: 1) 2006 – 2014; 2) 2014 – 2017; 3) 2017 – 2020; 4) 2020 - Now

Seasonal Trends Mostly Stayed the Same; Levels Changed

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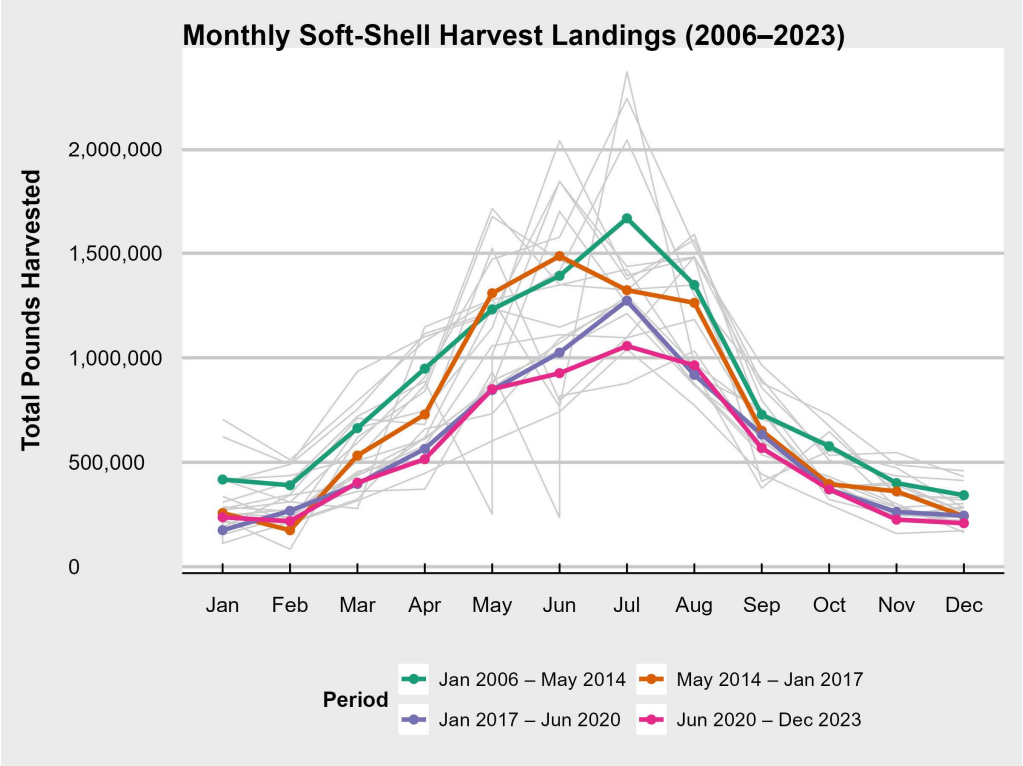
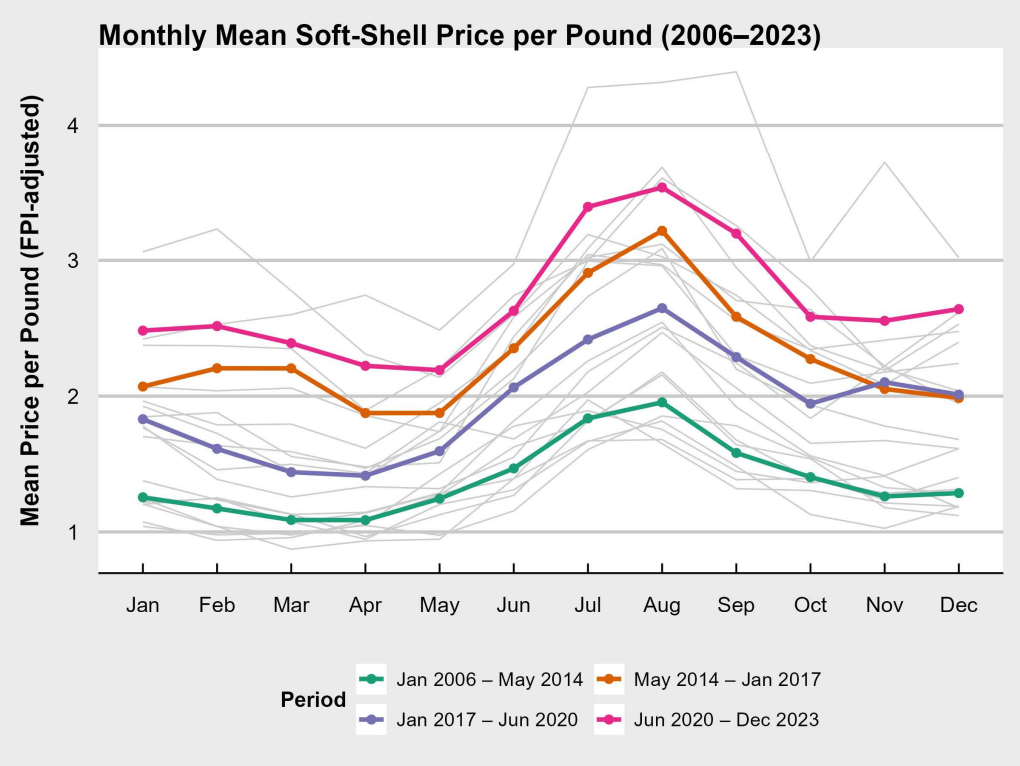
- Higher summer landings
- Peak in July, except for 2014 – 2017

- Similar seasonal pattern across periods
- High price through the summer of 2021, likely due to a post-pandemic tourism boom





1) 2006 – 2014; 2) 2014 – 2017; 3) 2017 – 2020; 4) 2020 - Now

Seasonality of Precipitation Closures

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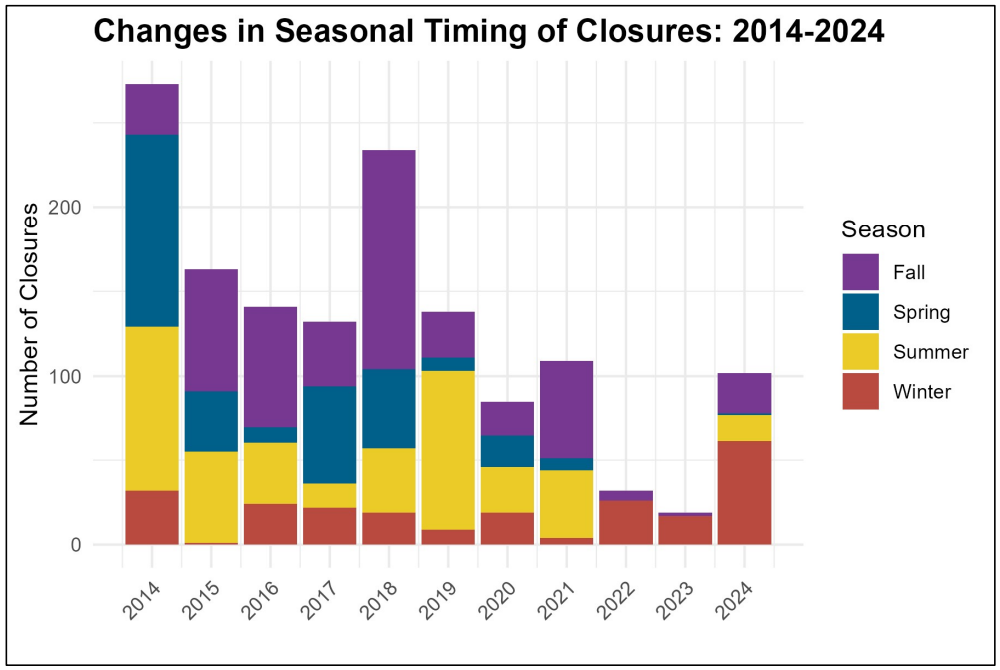
- GMRI intern Julia Victor examined precipitation closures in Maine from 2014-2024
  - 234 closures in 2018
  - 102 closures in 2024

- More recent closures have been in the Winter (December-February)
- Analysis found <u>no visible relationship</u> between closed days per year and annual soft-shell clam landings
- Further analysis on closure data could provide more insight on the effect precipitation has on landings



Conclusions

- While landings have been declining this past decade, prices have remained relatively stable
  - Price jumps in 2015 and 2021
  - COVID-19 pandemic impacts include more volatile prices, possible changes in demand, and coastal accessibility
  - Not all price shifts are due to changes in supply (i.e., lower landings)
- Various environmental changes have an impact on total landings in Maine
  - Cumulative impacts need to be investigated
  - Other impacts to consider as data become available: coastal temp change, ice cover

Technical Report

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Scan to explore the full report

Discover the latest research from GMRI's Coastal and Marine Economics Lab

Questions?
Find us at the poster session and at the GMRI table!

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